HR Analytics – Employee Attrition Prediction & Workforce Insights

Business Report

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Tools Used:

Python | MySQL | Power BI | Microsoft Excel | Machine Learning

Project Type:

End-to-End Data Analytics & Machine Learning Project

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Executive Summary

Employee attrition is one of the major challenges faced by organizations because it increases recruitment costs, affects workforce productivity, reduces employee morale, and impacts overall business performance. Understanding the factors that influence employee turnover enables organizations to develop effective retention strategies and improve workforce stability.

This project analyzes employee demographics, compensation, job satisfaction, work-life balance, career progression, and organizational factors to identify the key drivers of employee attrition. The project follows a complete analytics workflow, including data cleaning, exploratory data analysis (EDA), statistical analysis, feature engineering, feature selection, machine learning model development, SQL business analysis, and Power BI dashboard creation.

Multiple machine learning models were trained and evaluated to predict employee attrition, while SQL analysis and interactive dashboards provided valuable business insights to support data-driven HR decision-making.

Business Problem

Employee turnover has a significant financial and operational impact on organizations. High attrition leads to increased recruitment and training costs, loss of experienced employees, reduced productivity, and lower employee engagement.

HR departments often struggle to identify employees who are at risk of leaving and understand the factors contributing to attrition. Without proper data analysis, organizations cannot develop targeted retention strategies.

This project aims to analyze employee data, identify key attrition drivers, and build predictive models that help HR teams make informed decisions.

Project Objectives

- Clean and preprocess employee data.
- Perform Exploratory Data Analysis (EDA).
- Conduct statistical analysis.
- Apply feature engineering techniques.
- Perform feature selection.
- Build multiple machine learning classification models.
- Compare model performance.
- Perform SQL-based business analysis.
- Develop an interactive Power BI dashboard.
- Generate actionable business recommendations to improve employee retention.

Dataset Description

The dataset contains employee information collected from various HR-related attributes.

Dataset Summary

- Total Records: **1,470**
- Total Features: **35**
- Target Variable: **Attrition**
- File Format: CSV

Major Features

- Age
- Department
- Job Role
- Gender
- Monthly Income
- Business Travel
- Education
- Job Satisfaction

- Environment Satisfaction
- Work-Life Balance
- Overtime
- Years at Company
- Performance Rating
- Total Working Years
- Years Since Last Promotion

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- MySQL
- Power BI
- Git
- GitHub

Methodology

The project followed a structured analytics workflow:

1. Data Collection
2. Data Cleaning
3. Missing Value Handling
4. Exploratory Data Analysis
5. Statistical Analysis
6. Feature Engineering
7. Feature Selection
8. Machine Learning Model Development
9. Model Evaluation
10. SQL Business Analysis
11. Power BI Dashboard Development
12. Business Insights & Recommendations

Key Performance Indicators (KPIs)

- Total Employees
- Total Attrition
- Attrition Rate
- Average Monthly Income
- Average Age
- Average Years at Company
- Employees by Department
- Employees by Job Role
- Overtime Percentage
- Job Satisfaction Score
- Environment Satisfaction Score
- Work-Life Balance Score

Python Analysis

The Python workflow included:

- Data Cleaning
- Missing Value Handling
- Duplicate Removal
- Outlier Detection
- Exploratory Data Analysis
- Statistical Analysis
- Feature Engineering
- Feature Selection using Mutual Information and Recursive Feature Elimination (RFE)
- Machine Learning Model Development
- Hyperparameter Tuning
- Cross Validation
- Learning Curve Analysis
- Feature Importance Analysis

Machine Learning Models

The following classification algorithms were implemented:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Gradient Boosting Classifier
- K-Nearest Neighbors (KNN)

Model Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve
- Cross Validation
- Learning Curve

SQL Analysis

The SQL analysis included:

- Database Creation
- Table Design
- Data Validation
- Aggregate Functions
- GROUP BY Analysis
- CASE WHEN Statements
- Subqueries
- Correlated Subqueries
- Window Functions
- Advanced Business Queries
- Employee Demographics Analysis
- Salary Analysis
- Attrition Analysis
- Employee Satisfaction Analysis
- Career Growth Analysis

More than **60 business-oriented SQL queries** were developed to extract meaningful HR insights.

Power BI Dashboard

The interactive dashboard includes:

- Executive HR Dashboard
- Employee Demographics
- Department Analysis
- Job Role Analysis
- Attrition Analysis
- Salary Analysis
- Job Satisfaction Analysis
- Work-Life Balance Analysis
- Performance Analysis
- KPI Cards
- Interactive Filters & Slicers

Key Findings

- Employees working overtime exhibited a significantly higher attrition rate.
- Lower job satisfaction and poor work-life balance were strongly associated with employee turnover.
- Certain departments and job roles experienced higher attrition than others.
- Employees with shorter tenure contributed a large proportion of total attrition.
- Monthly income showed a positive relationship with employee retention.
- Career progression indicators, such as years since last promotion, influenced employee retention.
- Feature importance analysis identified the most influential variables affecting attrition.

Business Recommendations

- Improve work-life balance initiatives to reduce employee turnover.
- Monitor employees who frequently work overtime.
- Strengthen employee engagement and satisfaction programs.
- Develop transparent career progression and promotion plans.
- Review compensation strategies for high-risk employee groups.
- Implement early attrition prediction models to support proactive HR interventions.
- Continuously monitor workforce KPIs through interactive dashboards.

Conclusion

This project demonstrates a complete end-to-end HR Analytics solution by integrating Python, Machine Learning, SQL, and Power BI. The analysis successfully identified key factors influencing employee attrition and developed predictive models to assist HR teams in making informed decisions.

The project highlights practical skills in data cleaning, exploratory data analysis, statistical analysis, feature engineering, machine learning, SQL, business intelligence, and data visualization. The insights generated can help organizations reduce employee attrition, improve workforce planning, and support strategic HR decision-making.

Skills Demonstrated

- Data Cleaning
- Exploratory Data Analysis
- Statistical Analysis
- Feature Engineering
- Feature Selection
- Machine Learning
- Model Evaluation
- SQL
- Business Analytics
- Power BI
- Dashboard Development
- Data Visualization
- Git
- GitHub
